

Physical System



$$D = \kappa d$$

CP: z (surface potential)

CC: p (surface charge density)

input: $[D, v, b]$

Dimensionless Transformation & Preprocessing

Input vector: $[D, v, b]$

$$D \rightarrow \log_{10} D \rightarrow \text{z-score}$$

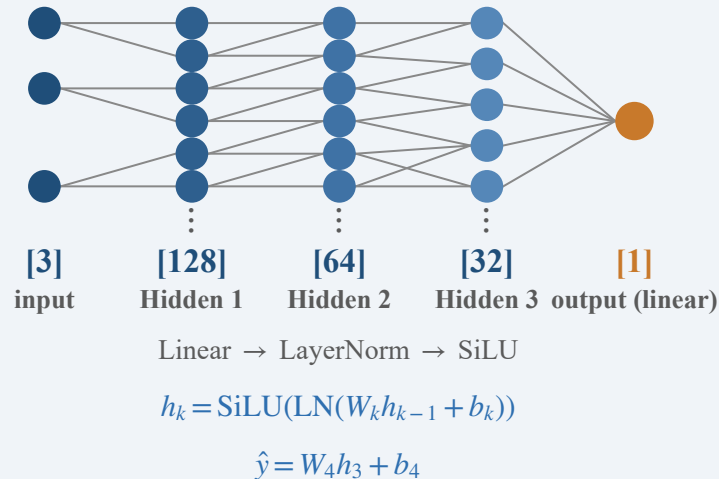
$$z \text{ (CP)} \rightarrow \text{linear} \rightarrow \text{z-score}$$

$$p \text{ (CC)} \rightarrow \log_{10} p \rightarrow \text{z-score}$$

$b \in \{0, 1\}$: boundary indicator (0: CP, 1: CC)

target: $u \rightarrow \log_{10} u$

Fully-Connected Network



Output

u
mid-plane potential

or
 E
repulsive energy
 $E = \int F(u) dD$

Architecture: $3 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$; each hidden layer: Linear $\rightarrow$ LayerNorm $\rightarrow$ SiLU; output: Linear (identity).

Trainable parameters: 10,881 (FC weights + biases) + 448 (LayerNorm γ, β) = 11,329.

Input ranges: $D \in [0.002, 10.5]$; $z \in [0.2, 10]$; $p \in [0.2, 5000]$; $u \in [0.003, 11.4]$; dataset: 64,000 samples; train/val/test = 80/10/10.